P1. A subgroup of index 2 is normal. For each element a, the conjugacy class $Cl(a) = \{ g^{-1}ag : g \in G \}$. It is clear that each conjugacy class either belongs to or disjoint with a fixed normal subgraph. Now let's focus on the structure of A_4.

$$A_4 = \{ e, \underbrace{\text{type } (123)}_{\binom{4}{3} \cdot 3!/3 = 8 \text{ elements}}, \underbrace{\text{type } (12)(34)}_{3 \text{ elements}} \}$$

Observe that $\sigma (a_1 a_2 \cdots a_k) \sigma^{-1} = (\sigma(a_1) \sigma(a_2) \cdots \sigma(a_k))$.

Hence $(12)(34)$, $(13)(24)$, $(14)(23)$ are mutually conjugate.

3-cycles (123), (142), (134), (243) are mutually conjugate;

and so are (132), (124), (143), (234).

Thus, the size of conjugacy classes are $1, 4, 4, 3$. It is impossible to assemble a group of 6.

P2. Commutative:

$$\begin{aligned}
&((g_1,h_1)(g_2,h_2))(g_3,h_3) \\
&= (g_1 \varphi_{h_1}(g_2), h_1h_2)(g_3,h_3) \\
&= (g_1 \varphi_{h_1}(g_2) \varphi_{h_1h_2}(g_3), h_1h_2h_3) \\
&= (g_1 \varphi_{h_1}(g_2) \varphi_{h_1}(\varphi_{h_2}(g_3)), h_1h_2h_3) \\
&= (g_1 \varphi_{h_1}(g_2 \varphi_{h_2}(g_3)), h_1h_2h_3) \\
&= (g_1, h_1) \cdot (g_2 \varphi_{h_2}(g_3), h_2h_3) \\
&= (g_1, h_1) \cdot ((g_2,h_2) \cdot (g_3,h_3)).
\end{aligned}$$

Identity: (e_G, e_H). Inverse: $(g,h)^{-1} = (\varphi_{h^{-1}}(g^{-1}), h^{-1})$

P3. ① f preserves group structure:

$$f(ab) = g^{-1}abg = g^{-1}agg^{-1}bg = f(a)f(b).$$

② f is injective: If $f(a)=f(b)$, then $g^{-1}ag = g^{-1}bg$.

Hence $\underbrace{g(g^{-1}ag)g^{-1}}_{= a} = \underbrace{g(g^{-1}bg)g^{-1}}_{= b}$.

③ f is surjective: $f(gag^{-1}) = a$.

In conclusion, f is an iso from G to itself.

Let $G = S_3$, $g = (12)$. Then $g^{-1}(123)g = (213) \neq (123)$. Hence, $f \neq id_G$.

P4. Clearly $\{(g, e_H) : g \in G\}$ is a subgroup of $G \times H$ isomorphic to G.

Similarly $\{(e_G, h) : h \in H\}$ is a subgroup of $G \times H$ isomorphic to H.

Therefore, if $G \times H$ is abelian, then G and H must be abelian.

If G & H are abelian, then $\forall g_1, g_2 \in G$, $h_1, h_2 \in H$,

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2) = (g_2g_1, h_2h_1) = (g_2, h_2)(g_1, h_1).$$

Hence, $G \times H$ is abelian.